



US012414244B2

(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 12,414,244 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **MAGNETICALLY MOUNTABLE CABLE HUB**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 449 days.

(21) Appl. No.: **17/882,279**

(22) Filed: **Aug. 5, 2022**

(65) **Prior Publication Data**

US 2024/0049405 A1 Feb. 8, 2024

(51) **Int. Cl.**
H05K 5/00 (2025.01)
G06F 1/16 (2006.01)
H05K 5/02 (2006.01)

(52) **U.S. Cl.**
CPC **H05K 5/0073** (2013.01); **G06F 1/1632** (2013.01); **H05K 5/0217** (2013.01); **H05K 5/0047** (2013.01)

(58) **Field of Classification Search**

CPC H05K 5/0073
See application file for complete search history.

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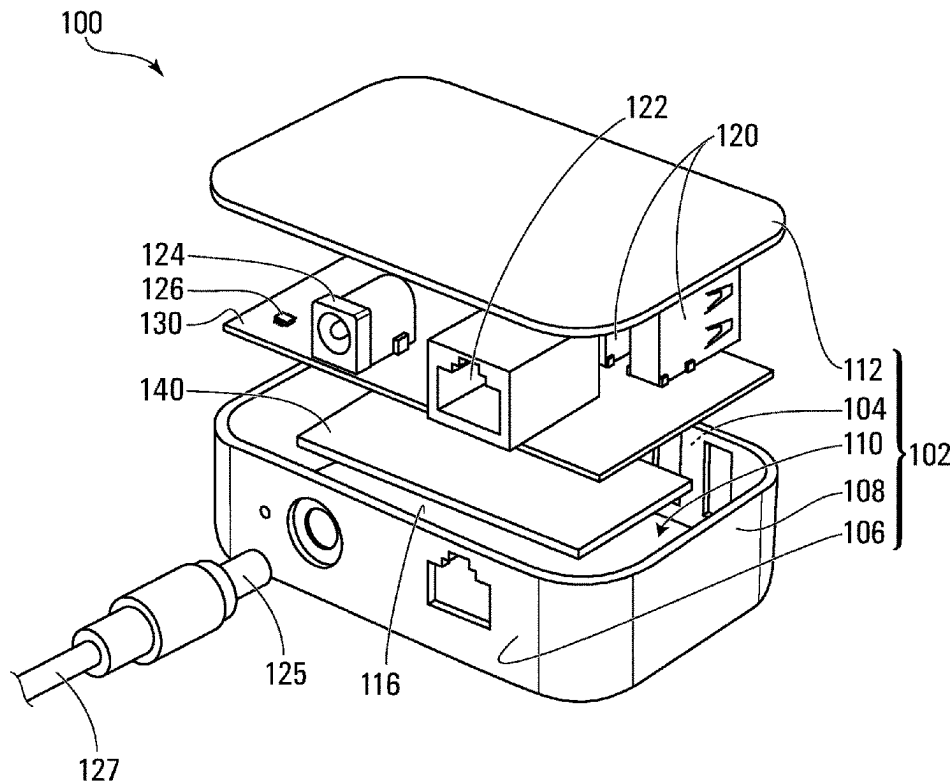
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(57) **ABSTRACT**

In one aspect, a cable hub includes a housing having a base and a wall. The base has an interior surface and an exterior surface. A plurality of cable ports is disposed in the wall. A mounting magnet is affixed to the interior surface of the base. The mounting magnet has a sufficient pull strength for magnetically mounting the cable hub to a ferromagnetic surface upon placement of the exterior surface of the base against the ferromagnetic surface. The cable hub may be a Universal Serial Bus (USB) hub. In another aspect, a kit includes the cable hub and at least one ferromagnetic plate. The ferromagnetic plate may be affixable to a surface using an adhesive material.

20 Claims, 10 Drawing Sheets



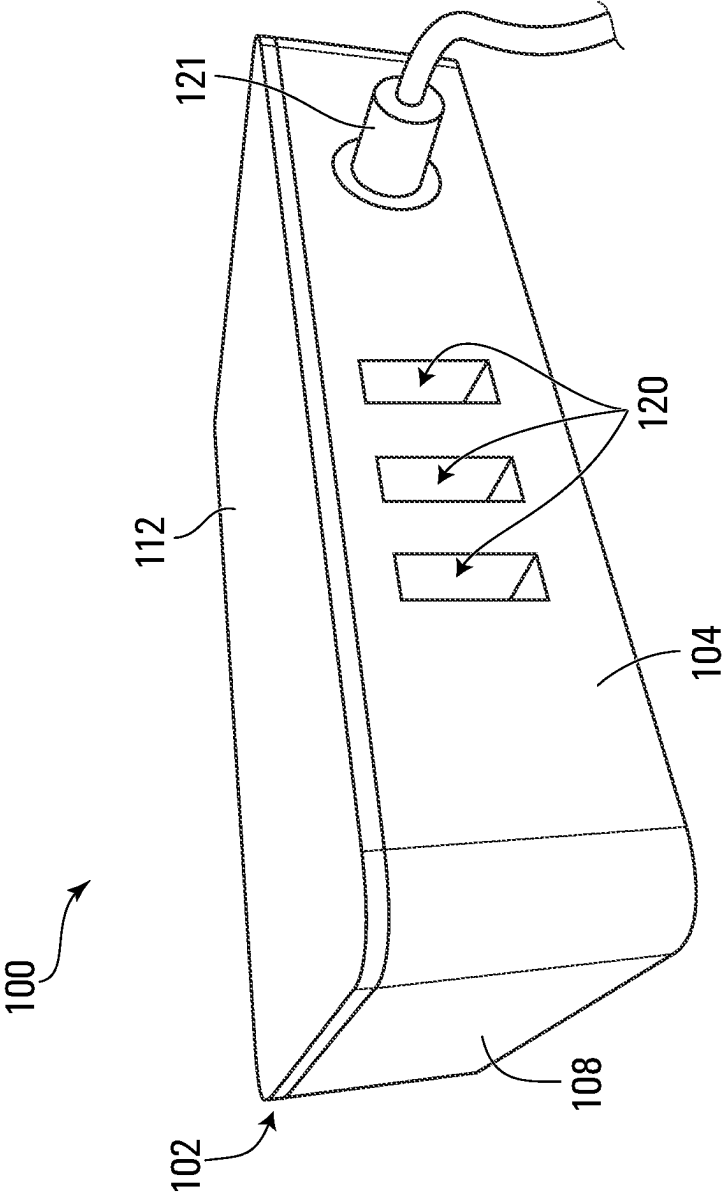


FIG. 1

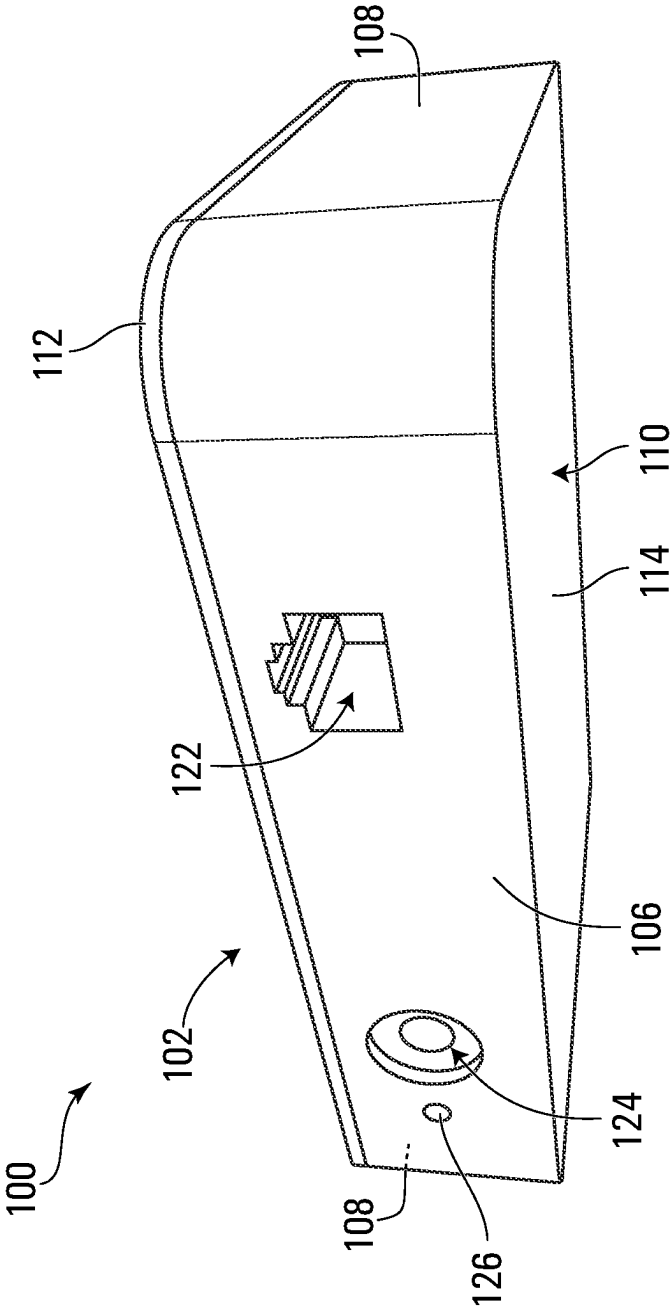


FIG. 2

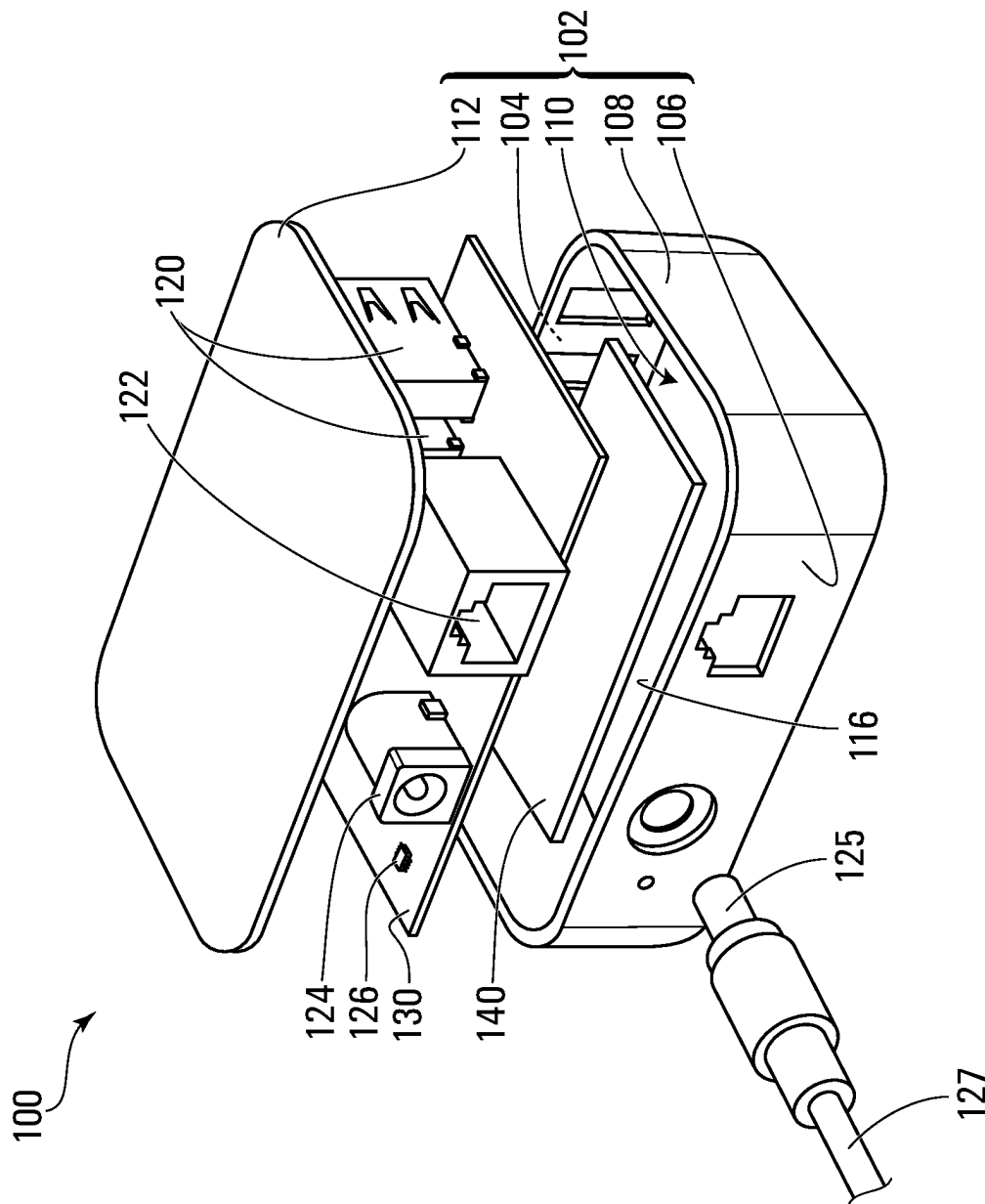


FIG. 3

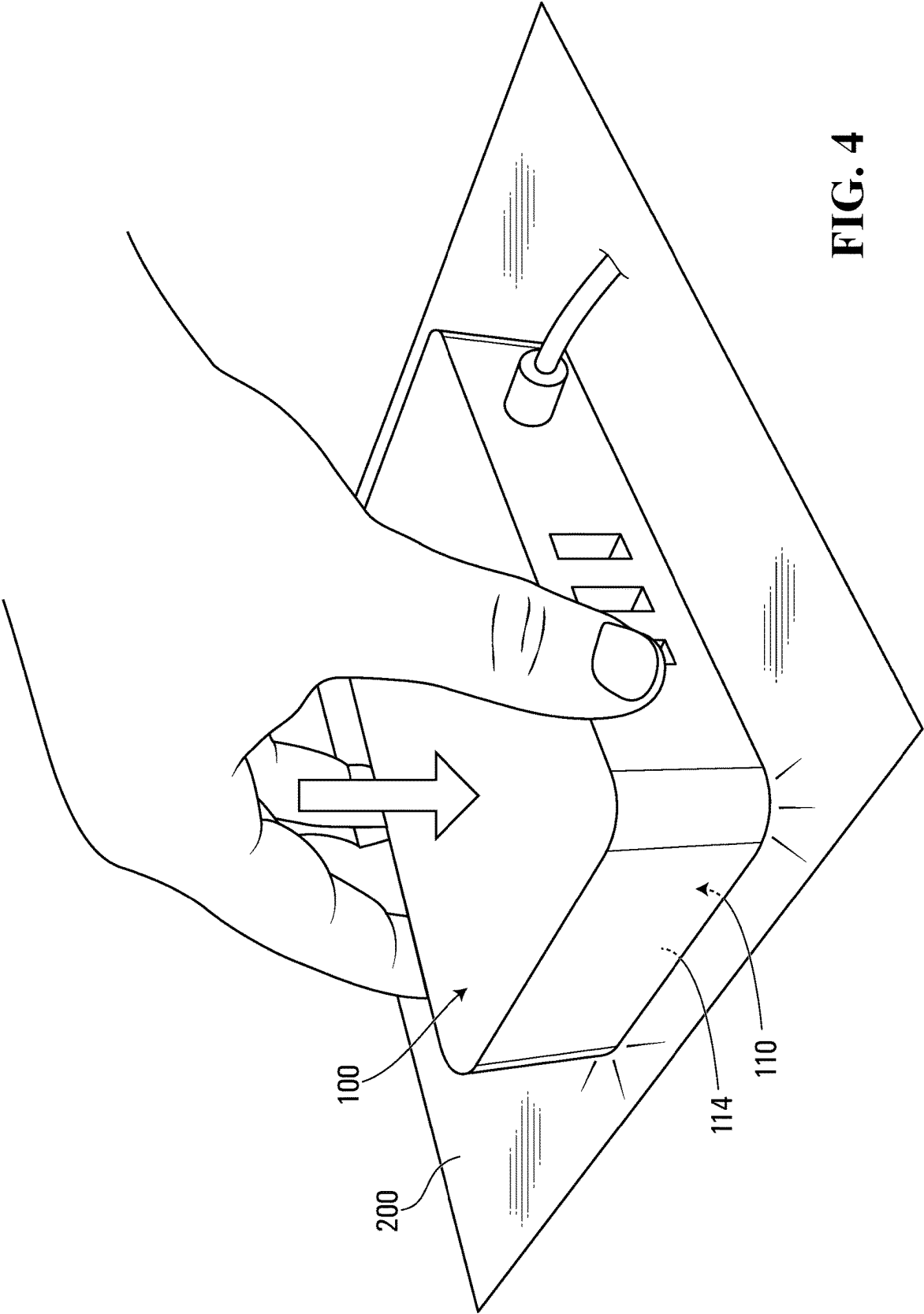


FIG. 4

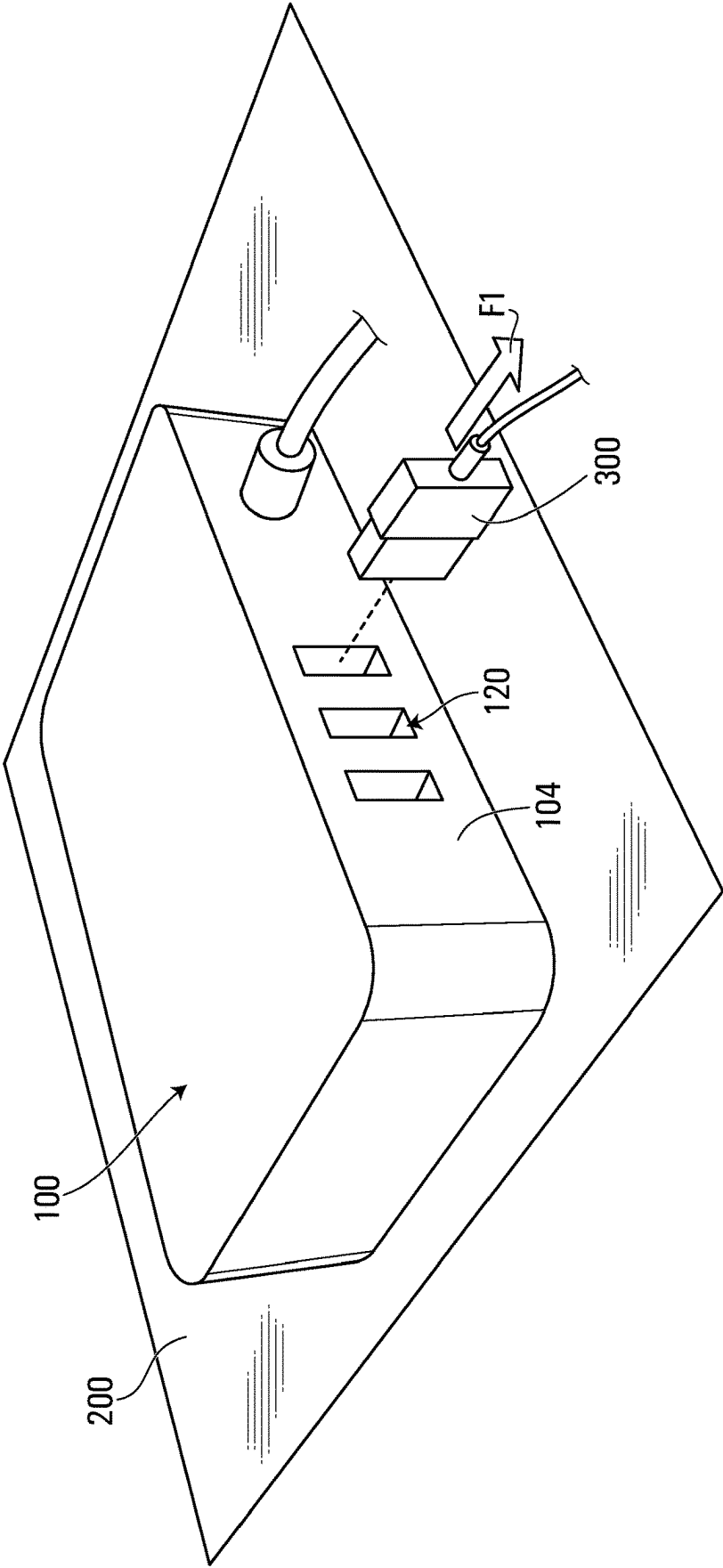


FIG. 5

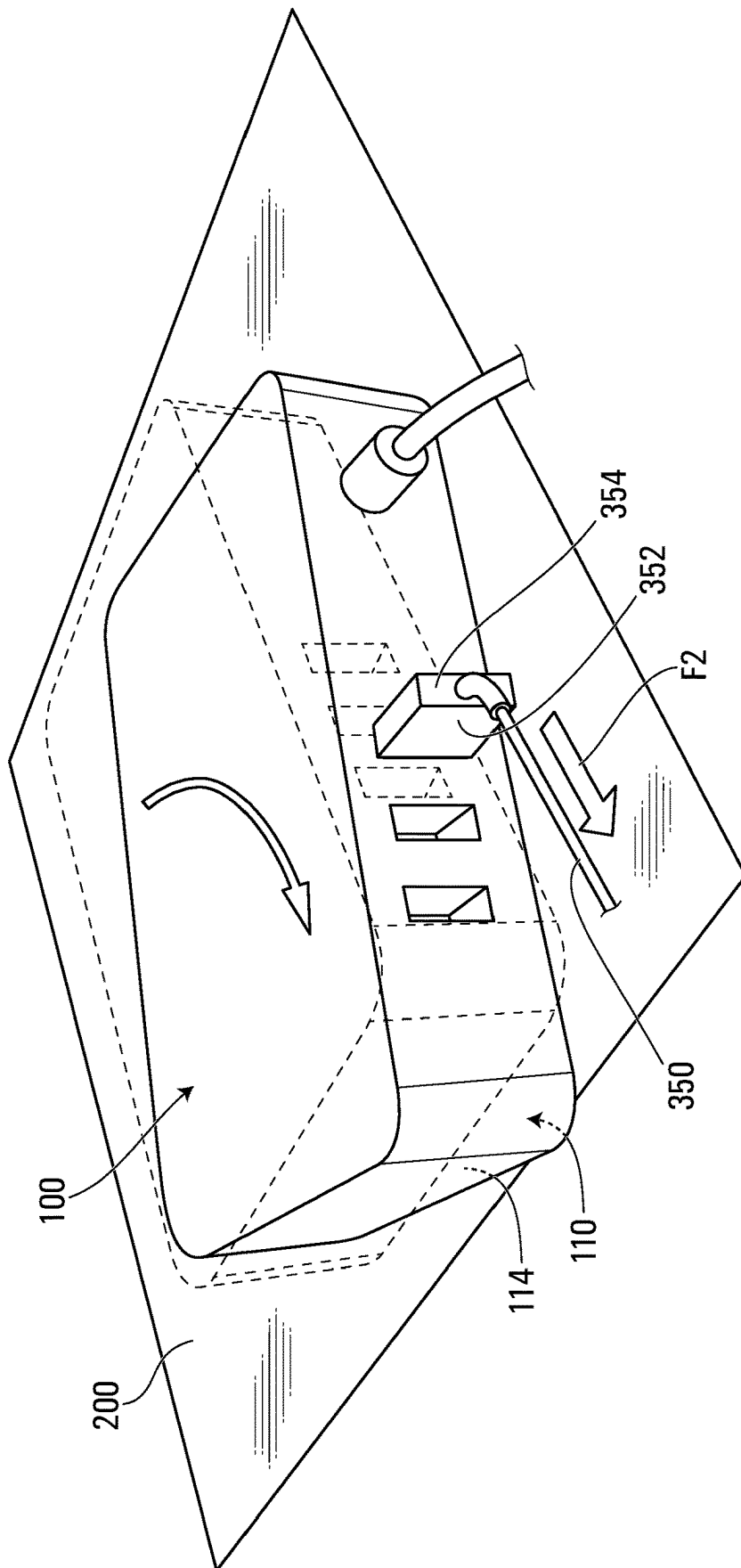


FIG. 6

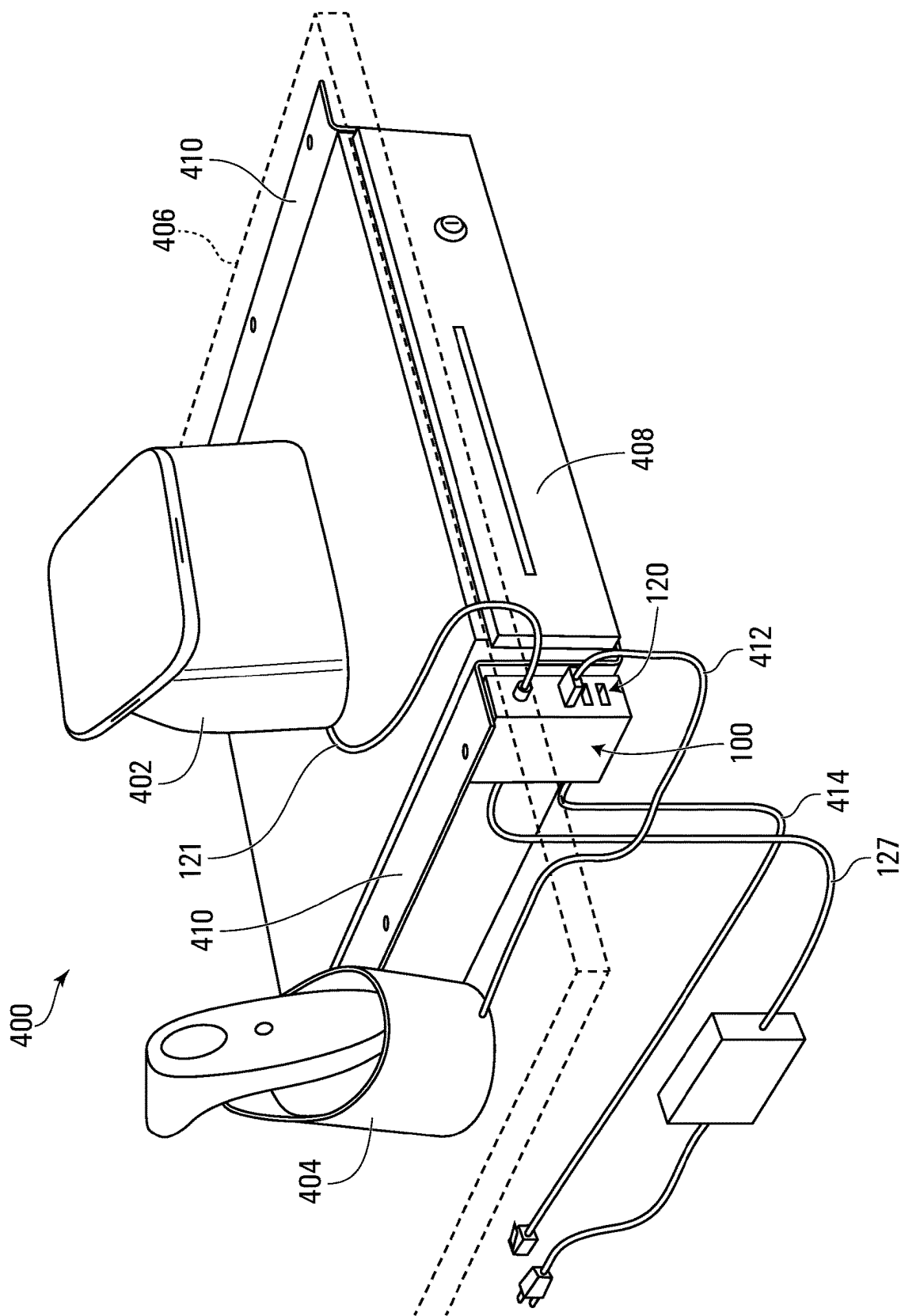


FIG. 7

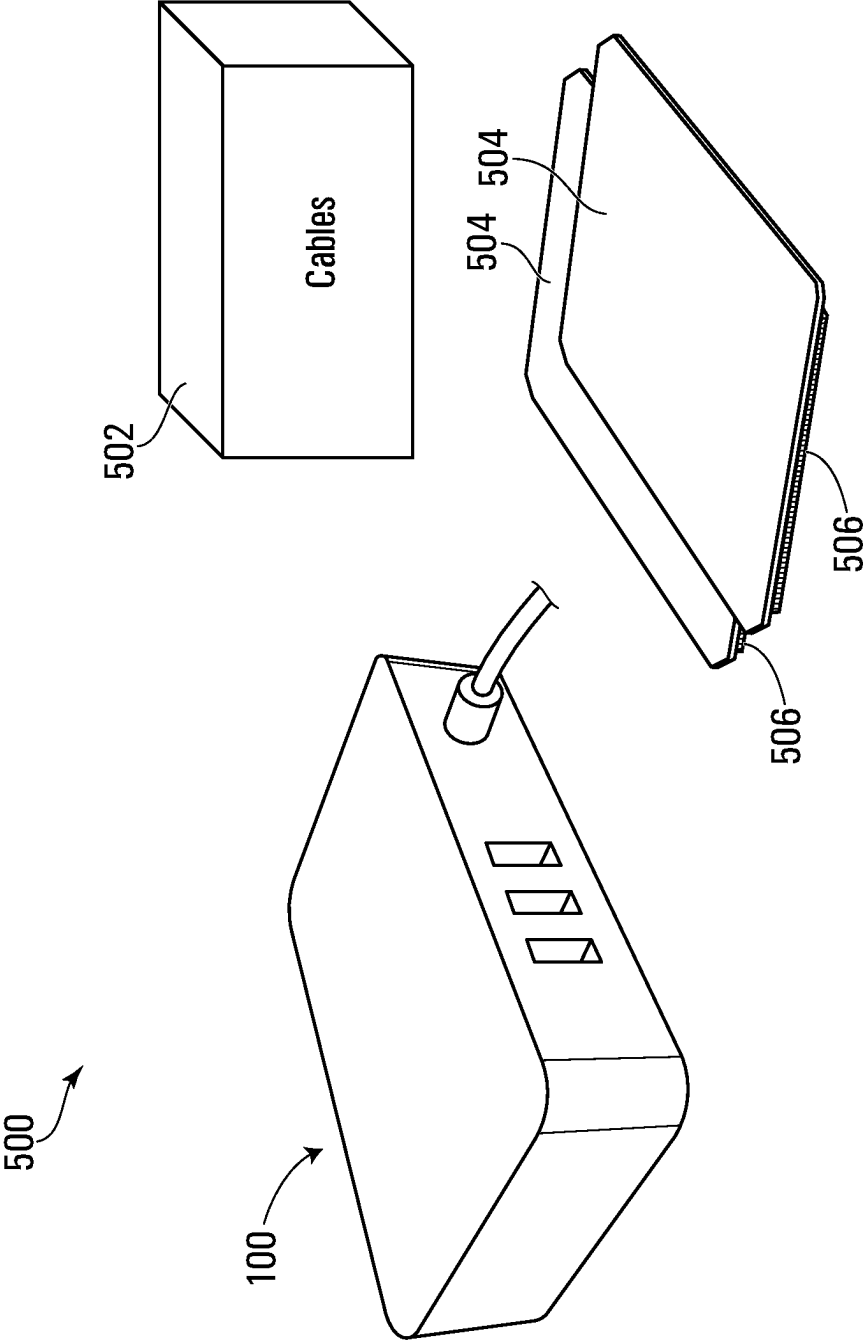


FIG. 8

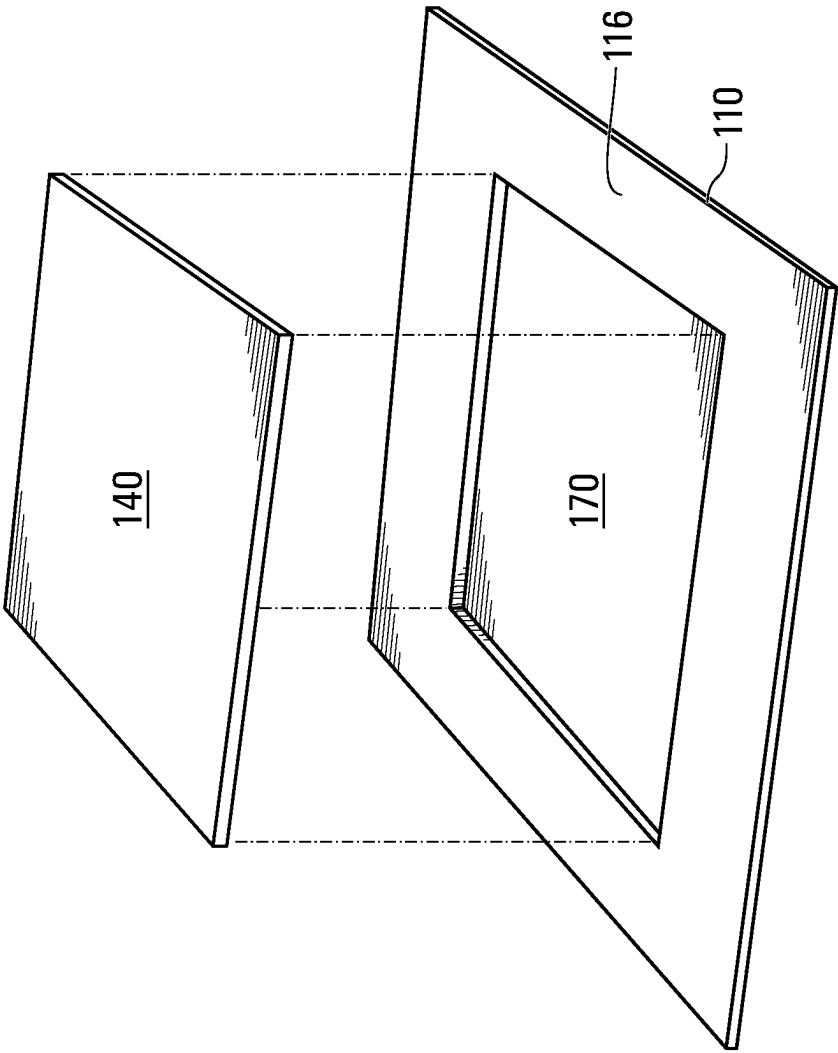


FIG. 9

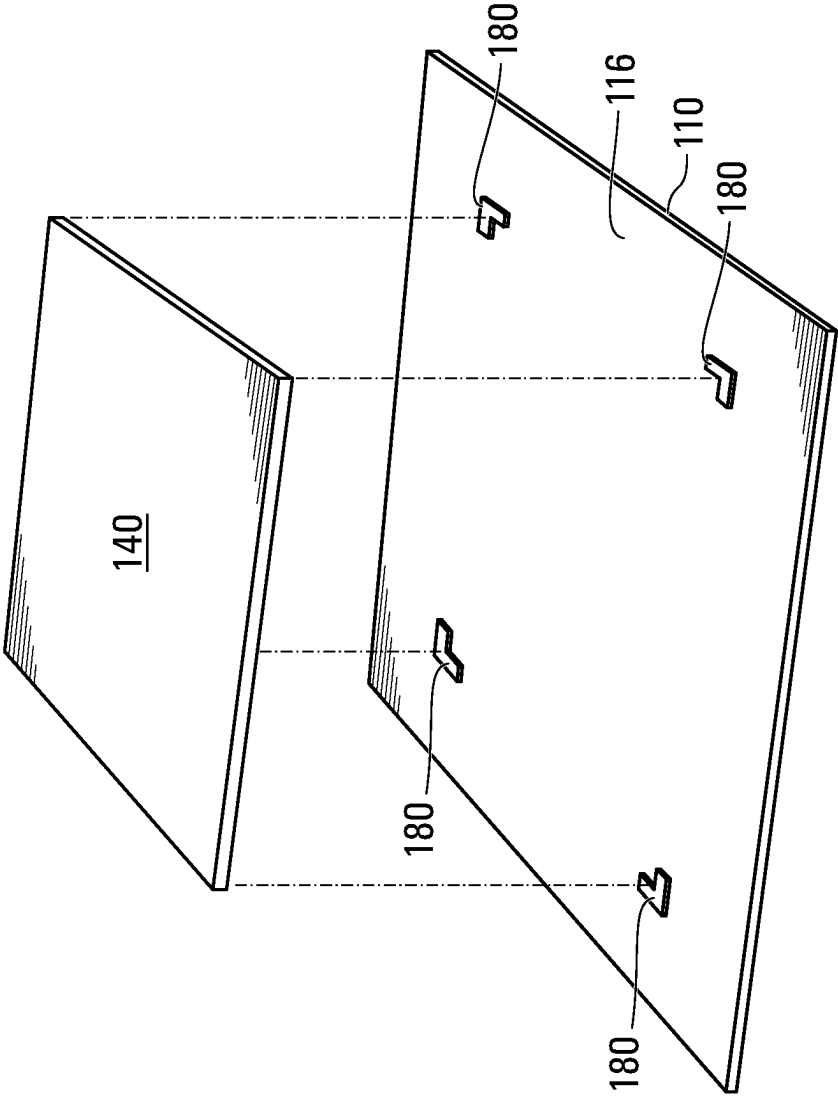


FIG. 10

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MAGNETICALLY MOUNTABLE CABLE HUB

TECHNICAL FIELD

The present disclosure relates to cable hubs, and more particularly to a magnetically mountable cable hub.

BACKGROUND

A Universal Serial Bus (USB) hub is an electronic device that can be connected to a host device (e.g., a computer) to increase the number of available USB ports by which peripheral electronic devices may be connected to the host device. In one configuration, the USB hub has a single “upstream-facing” USB port for connection with the host, e.g., via an insertable (male) USB connector, and multiple “downstream-facing” USB ports, each of which may be a (female) USB receptacle. Peripheral devices connected with the downstream-facing USB ports can intercommunicate with the host device via the upstream-facing USB port. The upstream-facing and downstream-facing USB ports may for example be USB Type A (USB-A) ports, USB Type C (USB-C) ports, other types of USB ports, or a combination of these.

A USB hub is one example of a cable hub. Other examples of cable hubs include an Ethernet™ hub, a laptop docking station, or a USB dongle.

SUMMARY

In one aspect, there is provided a cable hub comprising: a housing having a base and a wall, the base having an interior surface and an exterior surface; a plurality of cable ports disposed in the wall; and a mounting magnet affixed to the interior surface of the base, the mounting magnet having a sufficient pull strength for magnetically mounting the cable hub to a ferromagnetic surface upon placement of the exterior surface of the base against the ferromagnetic surface.

In some embodiments, the interior surface of the base defines a recess for the mounting magnet and the mounting magnet is received within the recess.

In some embodiments, the recess is a channel and the magnet is an elongate magnet.

In some embodiments, the mounting magnet is fixed to the interior surface of the base by a resilient adhesive.

In some embodiments, the interior surface of the base defines relief features abutting opposite sides of the mounting magnet.

In some embodiments, the cable hub is a Universal Serial Bus (USB) hub and each of the plurality of cable ports is a USB port.

In some embodiments, the pull strength of the mounting magnet and a coefficient of friction of the exterior surface of the base collectively promote a static force between the base and the ferromagnetic surface that exceeds a minimum necessary force, applied orthogonally to the wall, for connecting or disconnecting a friction-fit USB connector to or from, respectively, one of the USB ports.

In some embodiments, the USB port is a USB type A (USB-A) port and the USB connector is a USB-A connector.

In some embodiments, the USB port is a USB type C (USB-C) port and the USB connector is a USB-C connector.

In some embodiments, the housing is generally cuboid in shape, the wall is a first wall orthogonal to the base, and the cable hub further comprises a second wall opposite the first

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wall, the second wall having both of a network port and a power adapter socket disposed therein.

In another aspect, there is provided a kit comprising: a ferromagnetic plate; and

a cable hub including: a housing having a base and a wall, the base having an interior surface and an exterior surface; a plurality of cable ports disposed in the wall; and a mounting magnet fixed to the interior surface of the base, the mounting magnet having a sufficient pull strength for magnetically mounting the cable hub to the ferromagnetic plate upon affixing of the ferromagnetic plate to a surface and placement of the exterior surface of the base against the ferromagnetic plate.

Some embodiments of kit further comprise an adhesive material on one side of the ferromagnetic plate, the adhesive material configured for the affixing of the ferromagnetic plate to the surface.

In some embodiments, the ferromagnetic plate is a first ferromagnetic plate and the kit further comprises a second ferromagnetic plate.

In some embodiments, the cable hub is a Universal Serial Bus (USB) hub and each of the plurality of cable ports is a USB port.

BRIEF DESCRIPTION OF THE DRAWINGS

In the figures which illustrate example embodiments, FIG. 1 is a top front perspective view of a magnetically mountable cable hub;

FIG. 2 is a bottom rear perspective view of the cable hub of FIG. 1;

FIG. 3 is an exploded view of the cable hub of FIG. 1;

FIG. 4 depicts magnetic mounting of the cable hub of FIG. 1 to a ferromagnetic surface;

FIG. 5 depicts a first usage scenario of the magnetically mounted cable hub of FIG. 4;

FIG. 6 depicts a second usage scenario of the magnetically mounted cable hub of FIG. 4;

FIG. 7 illustrates an example use of the cable hub of FIG. 1 in a retail setting;

FIG. 8 illustrates a kit by which the magnetically mountable cable hub of FIG. 1 may be commercially sold;

FIG. 9 depicts an example recessed interior surface of a base of a housing of a cable hub configured to receive a mounting magnet; and

FIG. 10 depicts an example base of a housing of a cable hub having relief features configured to abut a mounting magnet on opposite sides.

DESCRIPTION

In this document, any use of the term “exemplary” should be understood to mean “an example of” and not necessarily to mean that the example is preferable or optimal in some way. Terms such as “top,” “bottom,” and “side” may be used to describe features of some embodiments in this description but should not be understood to necessarily connote an orientation of the embodiments during manufacture or use.

FIGS. 1 and 2 depict an example embodiment of a cable hub 100 in top front perspective view and bottom rear perspective view, respectively. FIG. 3 is an exploded view of the cable hub 100 of FIGS. 1 and 2. In the present embodiment, the cable hub 100 is a USB hub and may alternatively be referred to as such. In alternative embodiments, the cable hub may be something other than a USB hub, e.g., an Ethernet™ hub.

The cable hub **100** has a housing **102** made from a non-ferromagnetic material, such as Polycarbonate/Acrylonitrile-Butadiene-Styrene (PC/ABS). In the present embodiment, the housing **102** has a substantially cuboid shape with rounded corners. The housing **102** includes a front wall **104**, a rear wall **106**, two opposed side walls **108**, a base **110**, and a removable (e.g., snap-fit) cover **112**. The base **110** has an exterior surface **114** (FIG. 2) and an interior surface **116** (FIG. 3) and, in the present embodiment, is substantially flat. In alternative embodiments, the exterior surface of the base could be textured for grip or durability, corrugated (e.g., for airflow), and/or substantially concave in at least one dimension (e.g., for mounting to a cylindrical surface).

Three cable ports **120** are disposed in the front wall **104** (see FIGS. 1 and 3). In the present embodiment, the cable ports **120** are USB-A ports, and may thus be referred to as USB ports **120**. In alternative embodiments, the cable ports may be other types of USB ports or of non-USB cable ports and may number more than or less than three.

As best seen in FIG. 1, the front wall **104** has a fixed power and data cable **121** disposed therein. In this context, the term “fixed” refers to a non-removable, i.e., permanently attached, cable. In some embodiments, the permanent attachment may serve as an indicator that the cable **121** is an upstream-facing cable. In such embodiments, the cable ports **120**, to which data cables are removably connectable, may conversely be understood to be downstream facing. These distinctions may reduce a risk of cable hub **100** misconfiguration and/or a risk of cord tangling during storage or use. The power and data cable **121** is not necessarily fixed in all embodiments.

The power and data cable **121** may terminate with a male connector, such as a male USB-C plug. The cable **121** can be used to carry power and data signals to an interconnected device, such as a countertop tablet stand of a point-of-sale (POS) system with a complementary female receptacle (e.g., a USB-C receptacle), as will be described.

A possible rationale for locating (disposing) the power and data cable **121** in the front wall **104** of the cable hub **100** may be to facilitate reaching a counter-top POS device should the cable hub **100** be mounted under a counter with the rear wall **106** facing rearwardly.

Referring to FIGS. 2 and 3, a network port **122** is disposed in the rear wall **106** of the housing **102**. In the present embodiment, the network port **122** is local area network (LAN) port, and more particularly an RJ45 Ethernet™ port. The network port **122** serves as an interface between the cable hub **100** and a LAN and/or the Internet. Also disposed in rear wall **106** are a power port **124** (power adapter socket) and an indicator light **126**, which may be an LED. In the present embodiment, the power port **124** is a Direct Current (DC) power port designed to receive a barrel connector **125** of a power adapter cable.

A possible rationale for locating (disposing) the network port **122** in the rear wall **106** of the cable hub **100**, opposite from the front wall **104** in which the cable ports **120** are disposed, may be to reduce a likelihood that a network cable plugged into the network port **122** will interfere with the insertion/disconnection of cables into/from the cable ports **120** in the front wall **104**. Another possible rationale for disposing the network port **122** in the rear wall **106** may be to increase a likelihood that the network port **122** will face an RJ45 wall plate socket when the cable hub **100** has been mounted. For example, if the cable hub **100** is mounted under a desk or counter with its front wall **104** facing forward (e.g., for ease of access of cable ports **120**), then the

rear wall **106** will face rearwardly. In this orientation, the network port **122** may face a wall behind the desk or counter. Since RJ45 wall plate sockets are commonly installed in walls under desks or counters, interconnection of the cable hub **100** with the wall plate socket may be possible with a minimum length of network cable.

The cable ports **120**, power and data cable **121**, network port **122**, power port **124**, and indicator light **126** are all electrically and mechanically coupled (e.g., soldered) to a PCB **130** (FIG. 3) that is internal to the housing. In some embodiments, the PCB **130** may include circuitry (not illustrated) for controlling operation of the indicator light **126**. In one example, the circuitry may cause the indicator light **126** to illuminate steadily to indicate a power-on status of the cable hub **100**. In another example, the circuitry may cause the light **126** to blink in the event of a detected problem with power. In some embodiments, a separate indicator light associated with each cable port **120** (not depicted) may be used indicate, e.g., power or data flow or status for the associated cable port **120**.

Referring to FIG. 3, a mounting magnet **140** is affixed to the interior surface **116** of the base **110** of housing **102**. A purpose of the mounting magnet **140** is to permit the cable hub **100** to be easily attached to a ferromagnetic surface without tools. In the present embodiment, the mounting magnet **140** has a flattened cuboid shape and an extent that covers a majority of the interior surface **116** of the base **110**. The mounting magnet **140** may for example be a rare earth magnet.

In the present embodiment, the mounting magnet **140** is affixed to the interior surface **116** using an adhesive, such as glue. The adhesive may be resilient to help absorb or dissipate any shock that may be imparted upon the housing **102**, e.g., to minimize a risk of detachment of the mounting magnet **140** from the interior surface **116**. Affixing the mounting magnet **140** internally to the cable hub housing **102** may protect it from external forces that could otherwise result in magnet damage, such as magnet chipping (to which at least some rare earth magnets are susceptible). Internal mounting of the magnet **140** may also limit the amount of force required to detach the cable hub **100** from a ferromagnetic surface. Another potential benefit of internal magnet placement may be hiding the magnet from possible access by a child.

FIG. 4 depicts the manner in which a user may mount the cable hub **100** of FIG. 1 to a horizontally oriented ferromagnetic surface **200**. To mount the cable hub **100**, the user places the exterior surface **114** of the base **110** against the ferromagnetic surface **200**. The mounting magnet **140** magnetically attracts the ferromagnetic surface **200** through base **110**, thereby causing the cable hub **100** to become mounted to surface **200**. Conveniently, no tools are required to mount the cable hub **100**. Moreover, the user may easily adjust a position of the cable hub **110** on the ferromagnetic surface **200** without tools simply pulling the cable hub **100** away from the ferromagnetic surface and re-mounting as desired.

The exterior surface **114** of the base of the cable hub **100** may have a high coefficient of friction. Optionally, the pull strength of the mounting magnet **140** (i.e., the extent to which the magnet attracts the ferromagnetic surface **200**) and the coefficient of friction of the exterior base surface **114** may be collectively configured to promote the behaviors described below in the context of FIGS. 5 and 6.

FIG. 5 depicts a disconnection of a friction-fit USB connector **300** (a form of friction-fit cable connector) from one of the USB ports **120** of the cable hub **100**. In FIG. 5, the cable hub **100** has been magnetically mounted to ferro-

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magnetic surface **200**. A pulling force **F1** is applied to the USB connector **300** orthogonally to, and away from, the front wall **104**.

In this scenario, the cable hub **100** remains stationary during the disconnection of the USB connector **300**. The reason is that the static force resulting at least partly from the magnetic and friction forces collectively are stronger than the pulling force **F1**. Put another way, the pull strength of the mounting magnet **140** and a coefficient of friction of the exterior surface **114** of the base **110** collectively promote a static force between the base **110** and the ferromagnetic surface **200** that exceeds a minimum necessary force, applied orthogonally to the front wall **104**, for connecting or disconnecting a friction-fit USB connector **300** to or from, respectively, one of the USB cable ports **120**. The user may thus be able to conveniently disconnect the USB connector **300** from the cable hub **100** using just one hand, i.e., without having to immobilize the cable hub **100** with the other hand. Similarly, it may be possible to insert the USB connector **300** into the USB port **120** using only one hand, by applying a pushing force to the connector **300** that is of equal magnitude to force **F1** but in the opposite direction.

FIG. **6** depicts a second scenario in which a USB cable **350** whose USB connector **352** is plugged into a cable port **120** of the cable hub **100** is subjected to an impulse pulling force **F2**, e.g., due to inadvertent cable snagging. The pulling force is not orthogonal to the wall as in FIG. **5** but rather is at an acute angle relative to the wall and substantially parallel to the ferromagnetic surface **200** to which the cable hub **100** is mounted.

In the scenario depicted in FIG. **6**, the cable hub **100** absorbs some of the impulse force **F2** by rotating while remaining magnetically mounted to the ferromagnetic surface **200**. Rotation may occur even if the force **F2** is identical in magnitude to the force **F1** (FIG. **5**) that was insufficient for causing the cable hub **100** to slide (translate) along the ferromagnetic surface **200**. The reason is that, in this scenario, the force **F2** is applied to a distal end **354** of a rigid USB connector **352** protruding from the cable hub **100**. Due to torque, the USB connector **352** “magnifies” the impulse force **F2** proportionally to the protruding length of the USB connector **352**. The magnitude of the resultant magnified force **F2'** at the USB hub will be greater than the magnitude of the incident impulse force **F2**. The magnified force **F2'** exceeds a static friction between the cable hub **100** and the surface **200**, causing the cable hub **100** to rotate. In some scenarios, the force **F2** may cause both rotation and translation.

The rotation of the cable hub **100** may absorb or dissipate some of the impulse force **F2** that would otherwise have been applied to the cable port **120**, e.g., if the cable hub **100** were mechanically fixed to the ferromagnetic surface **200**. The force **F2** may be absorbed partly by the acceleration of the mass of the cable hub **100** from its initial stationary state and partly by friction of the exterior surface **114** of base **110** against the ferromagnetic surface **200**. As a result, a risk of damage to the USB cable port **120** from such impulse forces may be reduced in comparison to a fixedly mounted cable hub **100**. As well, whatever mechanical coupling may be used to attach the USB cable port **120** to the cable hub **100** (e.g., soldering of the cable port **120** to the PCB **130** and affixing of the PCB **130** to the housing **102**), may be subjected to less strain. This may in turn permit lighter or lesser amounts of mechanical coupling material(s) to be used to form the mechanical coupling, which may reduce the weight and/or cost of the cable hub **100**. In some embodiments, additional mechanical reinforcement may neverthe-

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less be used to lessen any possible strain upon PCB **130**. For example, the additional mechanical reinforcement may be used to connect cable port **120** directly to the housing **102**. Rotation may also reduce wear on the cables and/or connected downstream devices.

FIG. **7** depicts a possible use of the magnetically mountable cable hub **100** in a retail setting. In FIG. **7**, the cable hub **100** provides an easily repositionable electrical interface between an exemplary point-of-sale (POS) system **400** and one or more peripheral devices. The POS system **400** is usable by a merchant to receive payments and track sales at a retail location. The depicted POS system **400** includes a tablet stand **402** and a barcode scanner **404**. These components sit atop a counter **406**, which may be located at a checkout area of a store.

In FIG. **7**, the counter **406** is depicted in dashed lines to reveal under-counter components of the POS system **400**. The under-counter components include a metal money drawer **408**. In the present embodiment, the money drawer **408** is attached to the underside of the counter **406** by metal brackets **410**. It will be appreciated that the money drawer **408** and the brackets **410** are made of metal for security reasons.

In the retail setting depicted in FIG. **7**, the cable hub **100** may be conveniently magnetically mounted near (or directly to) the money drawer **408**, e.g., to one of the brackets **410** by which the money drawer **408** is mounted to the underside of counter **406**. It will be appreciated that, in this example, the ferromagnetic surface **410** to which the cable hub **100** is magnetically mounted is oriented vertically. In this embodiment, the pull strength of the mounting magnet **140** (i.e., the extent to which the magnet **140** attracts the ferromagnetic surface **410**) is sufficient to support the weight of the cable hub **100** in a vertical orientation.

The countertop components of the POS system **400** are each connected (communicatively coupled) to the cable hub **100** by a respective cable. In particular, the countertop tablet stand **402** is connected to cable hub **100** via a power and data cable **121**. The barcode scanner **404** is connected to the cable hub **100** via USB cable **412**. Under the counter **406**, a power adapter cable **127** supplies power to the cable hub **100**. A network cable **414** (e.g., a Category 6 cable terminated by RJ45 connectors) communicatively couples the cable hub **100**, and thus the POS system **400**, to a LAN and/or the Internet.

It will be appreciated that magnetic mounting of the cable hub **100** below counter level as shown in FIG. **7** may allow many of the cables **127**, **412**, and **414** associated with the POS system **400** to be placed underneath the counter **406**. This may reduce the number of countertop cables, e.g., in comparison to tabletop USB hub placement, or in comparison to a configuration in which peripherals are connected to, e.g., spare USB ports of the tablet stand **402**. The risk of countertop cable snagging may accordingly be reduced. Moreover, because the cable hub **100** may be able to absorb impulse forces applied to connected USB cables in certain directions (in comparison to permanently mounted cable hubs, as described above in connection with FIG. **6**), a risk of damage to cable ports **120** may be limited.

If the merchant wishes to add or remove USB-connectible peripheral devices to or from the POS system **400** over time, e.g., as the needs of the merchant change, the cable hub **100** may be temporarily detached, without tools, to facilitate disconnection/connection of USB cables. When the desired cable connections have been made, the cable hub **100** may conveniently be magnetically re-mounted to the bracket **210**

in the same location, or possibly in a new location, e.g., if new cable lengths warrant a change in cable hub location.

FIG. 8 depicts a kit 500 by which the cable hub 100 may be commercially sold. The example kit 500 includes one cable hub 100, one or more cables, e.g., a power adapter cable and one or more USB cables (depicted inside a box 502 in FIG. 8), and a pair of ferromagnetic plates 504. In this example, each ferromagnetic plate 504 has an adhesive material 506 on one side, such as double-sided mounting tape. The plates 504 may allow the cable hub 100 to be used in settings that otherwise lack ferromagnetic surfaces.

Installation may involve affixing each plate 504 to a surface at a desired installation location using the adhesive material 506. The cable hub 100 may then be selectively magnetically mounted to either one or the other of the plates 504 as needed. The number of plates 504 may be greater or less than two in alternative embodiments of kit 500. In alternative embodiments, the plate(s) 504 may be affixed using mechanisms other than an adhesive material.

Various alternative embodiments are possible.

In some embodiments, the mounting magnet of a cable hub 100 may be received in a recess or channel defined in the interior surface of the base of the housing. An example is depicted in FIG. 9.

FIG. 9 depicts a base portion 110 of the cable hub 100 of FIGS. 1-3 in isolation from the remainder of the housing 102. The interior surface 116 of the base 110 has a recess 170 formed therein. In the depicted example, the recess 170 is sized to snugly receive the flattened cuboid-shaped mounting magnet 140. The relative sizing of the recess and the mounting magnet 140 may help to fix the position of the mounting magnet 140 relative to the base 110. An adhesive (not depicted) such as glue may be applied to help affix the mounting magnet 140 to the base 110 within the recess 170. In some embodiments, the mounting magnet may be elongate, and the recess may be a channel.

In some embodiments, the mounting magnet may be abutted on opposite sides by relief features defined on the interior surface of the base. An example is depicted in FIG. 10.

FIG. 10 depicts a base portion 110 of the cable hub 100 of FIGS. 1-3 in isolation from the remainder of the housing 102. The interior surface 116 of the base 110 has relief features 180 defined thereupon. In the depicted example, the relief features are L-shaped protuberances. The protuberances may be integrally formed with the base 110, e.g., by way of a molding process by which base 110 may be formed.

In the present embodiment, each of the L-shaped relief features 180 is configured to receive a respective corner of the mounting magnet 140 and abuts the corner from two sides. Collectively, the relief features 180 abut the mounting magnet 140 on opposite sides. This may help to fix the mounting magnet 140 relative to the base 110. An adhesive (not depicted) such as glue may be applied to help affix the mounting magnet 140 to the interior surface 116. In alternative embodiments, the relief features could have other shapes, such as blocks or cylindrical posts. Relief features, if present, may abut the mounting magnet at any location (not necessarily just at the corners of the magnet).

It will be appreciated that the features disclosed in the context of FIGS. 9 and 10, e.g., recesses or channels defined in, and/or relief features defined on, the interior surface 116 of the base, may be combined with features of the cable hub 100 disclosed herein, e.g., as shown in FIGS. 1 to 6 and as described hereinabove.

In alternative embodiments, the mounting magnet may be shaped differently from what is shown in FIGS. 3, 9, and 10.

For example, the mounting magnet may be bar-shaped or disk-shaped. In some embodiments, multiple mounting magnets may be used per cable hub 100. Each mounting magnet may be disposed at a different location on the interior surface 116 of the base 110 of housing 102.

In some embodiments, the pull strength of the magnet may be sufficient to support the weight of the cable hub 100 regardless of an orientation of the ferromagnetic surface. For example, the mounting magnet 140 may have a pull strength that is sufficient to magnetically mount the cable hub 100 to a horizontal ferromagnetic surface in an upside-down orientation, i.e., with base 110 facing up and cover 112 facing down.

In some embodiments, the mounting magnet 140 may be translatable, or otherwise movable, towards or away from the base 110, e.g., using one or more leadscrews. This may permit a degree to which the cable hub 100 is attracted to the ferromagnetic surface 200 to be adjusted. In some embodiments, the cable hub 100 may incorporate a microcontroller or CPU communicatively coupled to a force sensor and one or more actuators configured to move the magnet, e.g., via the leadscrew(s). The microcontroller or CPU may be operable to detect, via the force sensor, a current degree of force by which the cable hub 100 is held to the ferromagnetic surface 200. The microcontroller or CPU may be further operable to dynamically adjust the current degree of force by suitably moving the mounting magnet 140.

In at least some embodiments described above, the cable hub 100 incorporates a single PCB 130. In alternative embodiments, multiple interconnected PCBs or other circuitry mounting means may instead be used.

What is claimed is:

1. A cable hub comprising:

a housing having a base and a wall, the base having an interior surface and an exterior surface;

a plurality of cable ports in the wall; and

a mounting magnet affixed to the interior surface of the base, the mounting magnet having a sufficient pull strength for magnetically mounting the cable hub to a ferromagnetic surface upon placement of the exterior surface of the base against the ferromagnetic surface, the mounting magnet enclosed by the housing when the cable hub is magnetically mounted to the ferromagnetic surface.

2. The cable hub of claim 1, wherein the interior surface of the base defines a recess for the mounting magnet and wherein the mounting magnet is received within the recess.

3. The cable hub of claim 2, wherein the recess is a channel and wherein the magnet is an elongate magnet.

4. The cable hub of claim 1, wherein the mounting magnet is fixed to the interior surface of the base by a resilient adhesive.

5. The cable hub of claim 1, wherein the cable hub is a Universal Serial Bus (USB) hub and wherein each of the plurality of cable ports is a USB port.

6. The cable hub of claim 5, wherein the pull strength of the mounting magnet and a coefficient of friction of the exterior surface of the base collectively promote a static force between the base and the ferromagnetic surface that exceeds a minimum necessary force, applied orthogonally to the wall, for connecting or disconnecting a friction-fit USB connector to or from, respectively, one of the USB ports.

7. The cable hub of claim 6, wherein the USB port is a USB type A (USB-A) port and the USB connector is a USB-A connector.

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8. The cable hub of claim 6, wherein the USB port is a USB type C (USB-C) port and the USB connector is a USB-C connector.

9. The cable hub of claim 1, wherein the housing is generally cuboid in shape, wherein the wall is a first wall orthogonal to the base, and further including a second wall opposite the first wall, the second wall having both of a network port and a power adapter socket therein.

10. A kit comprising:

a ferromagnetic plate; and

a cable hub including:

a housing having a base and a wall, the base having an interior surface and an exterior surface;

a plurality of cable ports disposed in the wall; and

a mounting magnet fixed to the interior surface of the base, the mounting magnet having a sufficient pull strength for magnetically mounting the cable hub to the ferromagnetic plate upon affixing of the ferromagnetic plate to a surface and placement of the exterior surface of the base against the ferromagnetic plate, the mounting magnet enclosed by the housing when the cable hub is magnetically mounted to the ferromagnetic plate.

11. The kit of claim 10, further comprising including an adhesive material on one side of the ferromagnetic plate, the adhesive material configured for the affixing of the ferromagnetic plate to the surface.

12. The kit of claim 10, wherein the ferromagnetic plate is a first ferromagnetic plate and further including a second ferromagnetic plate.

13. The kit of claim 10, wherein the cable hub is a Universal Serial Bus (USB) hub and wherein each of the plurality of cable ports is a USB port.

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14. A cable hub comprising:

a housing having a base and a wall, the base having an interior surface and an exterior surface;

a plurality of cable ports in the wall; and

a mounting magnet affixed to the interior surface of the base, the mounting magnet having a sufficient pull strength for magnetically mounting the cable hub to a ferromagnetic surface upon placement of the exterior surface of the base against the ferromagnetic surface, the interior surface of the base defining relief features abutting opposite sides of the mounting magnet.

15. The cable hub of claim 14, wherein the mounting magnet enclosed by the housing when the cable hub is magnetically mounted to the ferromagnetic surface.

16. The cable hub of claim 14, further including a network port, the network port different than the plurality of cable ports.

17. The cable hub of claim 16, wherein the wall is a first wall and further including a second wall opposite the first wall, the network port in the second wall.

18. The cable hub of claim 14, further including a power port to receive a power adapter cable.

19. The cable hub of claim 14, further including a light emitting diode visible via the housing.

20. The cable hub of claim 14, wherein the relief features include a first relief feature, the first relief feature having a first portion and a second portion, the first portion of the first relief feature abutting a first portion of the mounting magnet and the second portion of the first relief feature abutting a second portion of the mounting magnet.

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